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Appointments:

2019 - Present Associate Professor, University of Massachusetts Boston
2012 - Present Assistant Professor, University of Massachusetts Boston
2010 - 2012 Postdoctoral Fellow, National Center for Ecological Analysis and Synthesis
2008 - 2010 Postdoctoral Fellow, Santa Barbara Long Term Ecological Research Project

Education:

2002-2008 UC Davis, Population Biology, M.S. 2003, Ph.D. 2008
1997-2001 Brown University, Bachelor of Science in Biology

Publications:

* - *Denotes undergraduate author,* † *- Denotes graduate student trainee,* ‡ *- Denotes postdoctoral trainee*

In Review

6. Dee, L. E., Ferraro, P. J., Siegel, K. J., Miller, S. J., **Byrnes, J. E. K.,** Correia, H. E., McIntosh, T. L., and S. Arif. In Revision. Seven myths about causal inference in ecology: clarifications and common ground.

5. Balasubramanian, S. ‡, Borja, C., Murillo, A. C. Wilkes, A. L.†, McPherson‡, M. L. Krumhansl, K. A., Dijkstra, J. A., and **J. E. K. Byrnes.** 2026. Comparative Assessment of Deep Learning Architectures for Underwater Subsurface Kelp Forest Segmentation with The Kelp-o-Tron. arXiv. https://arxiv.org/abs/2608.24594

4. Borja, C., B. A. McCollum†, **J. E. Byrnes**, K. Sebens, and A. C. Murillo. 2026. Leveraging existing sparse point annotations for benthic imagery dense segmentation. Third Workshop on Computer Vision for Ecology. arXiv. https://arxiv.org/html/2608.17561v1

3. Beaulieu, J., N. Knight, A. Hesketh, J. Lawlor, M. Gheţa, T. Peller, J. B. Burant, L. Super, E. Bledsoe, K. Benes, J. Dijkstra, and **J. E. K. Byrnes.** 2026. In Review. Historical subtidal regime shifts echoed in adjacent intertidal community. EcoEvoRxiv. https://ecoevorxiv.org/repository/view/12348/

2. Lawlor, J., A. Hesketh, J. Beaulieu, N. Knight, T. Peller, L. Super, A. Pereira, E. Bledsoe, J. B. Burant, J. Dijkstra, K. Benes, and **J. E. K. Byrnes**. 2026. In Review. Global Patterns Predict Local Biodiversity Shifts in a Climate Change Hotspot. EcoEvoRxiv. https://ecoevorxiv.org/repository/view/11898/

1. McCollum, B. A.†, **J. E. K. Byrnes**, and K. P. Sebens. 2026. In Review. Changes in species composition of sessile communities on subtidal rock walls in the southern Gulf of Maine during four decades of warming. bioRxiv. https://doi.org/10.64898/2026.03.01.708879

Refereed Papers

70. Beckley, B. A., A. Rassweiler, M. C. N. Castorani, **J. E. K. Byrnes**, D. C. Reed, and R. J. Miller. 2026. Direct and indirect effects of disturbance on net primary production in giant kelp forests. Ecology 107:e70393.

69. Dafforn, K. A., M. Borrelli, K. Bosma, **J. E. K. Byrnes**, L. Castro-Diaz, R. F. Chen, J. Christo, D. Cook, F. Peri, R. Shoer, M. Albert, K. Bradley, L. Firth, D. Kent, J. Knisel, D. Lopez†, C. R. Morris†, A. Novelly, L. Nurse, C. Osgood, S. Plesset, E. Solomon, C. Thornber, M. T. Sosa†, E. Zlevor, and P. Kirshen. 2026. A climate services organization supporting coastal resilience: lessons from the Stone Living Lab. Climatic Change 179:150.

68. Correia, H. E., L. E. Dee, **J. E. K. Byrnes**, J. R. Fieberg, M.-J. Fortin, C. Glymour, J. Runge, B. Shipley, I. Shpitser, K. J. Siegel, G. Sugihara, B. von Holle, and P. J. Ferraro. 2026. Best practices for moving from correlation to causation in ecological research. Nature Communications 17:1981. https://doi.org/10.1038/s41467-026-69878-z

67. von Schuckmann, K., A. Godoy-Faundez, V. Garçon, F. E. Muller-Karger, K. Evans, W. Appeltans, N. Bax, L. Benedetti Cecchi, A. Bernard, K. Bernard, **J. Byrnes,** G. Canonico, L. Chapuis, M. R. Clark, A. M. Darnaude, C. Davies, P. Englyst, A. Fransson, S. Hallam, E. Heslop, E. Holland, M. Hood, S. Kern, A. Liné, A. Lara-Lopez, N. Loose, B. Martín Míguez, C. R. McMahon, L. M. Nordlund, J. Post, S. Speich, A. Sutton, T. Tanhua, M. Telszewski, D. Poursanidis, and W. Yu. 2026. Global ocean indicators: Marking pathways at the science-policy nexus. *Marine Policy*. 184: 106922. https://www.sciencedirect.com/science/article/pii/S0308597X25003380

66. Dudney, J., L. E. Dee, R. Heilmayr, **J. Byrnes**, and K. Siegel. 2025. A Causal Inference Framework for Climate Change Attribution in Ecology. *Ecology Letters* 28:e70192. https://onlinelibrary.wiley.com/doi/10.1111/ele.70192

65. Eger, A. M., G. V. Wood, and **J. Byrnes.** 2025. An Environmental Niche Exploration Tool for Kelp Forest Management. *Ecology and Evolution* 15:e71459. https://onlinelibrary.wiley.com/doi/10.1002/ece3.71459

64. Pérez-Matus, A., F. Micheli, B. Konar, N. Shears, N. H. N. Low, D. K. Okamoto, T. Wernberg, K. A. Krumhansl, S. D. Ling, M. Kingsford, T. Navarrete-Fernandez, C. S. Ruz, and **J. E. K. Byrnes**. 2025. Kelp forests as nursery and foundational habitat for reef fishes. Ecology 106:e70007. https://esajournals.onlinelibrary.wiley.com/doi/full/10.1002/ecy.70007

63. Cavanaugh, K.C., Bell, T.W., Aerni, K.E., **Byrnes, J.E.K**., McCammon, S. & Smith, M.M. 2025. New Technologies for Monitoring Coastal Ecosystem Dynamics. *Annual Review of Marine Science*, 17, 409–433. https://www.annualreviews.org/content/journals/10.1146/annurev-marine-040523-020221

62. **Byrnes, J.E.K.** & Dee, L.E. 2025. Causal Inference With Observational Data and Unobserved Confounding Variables. *Ecology Letters*, 28, e70023. https://onlinelibrary.wiley.com/doi/full/10.1111/ele.70023

61. Wilkes, L. N.†, Barner, A. K., Keyes, A. A., Morton, D., **Byrnes, J. E. K.,** & Dee, L. E. 2024. Quantifying co-extinctions and ecosystem service vulnerability in coastal ecosystems experiencing climate warming. Global Change Biology. 30, e17422. https://doi.org/10.1111/gcb.17422

60. Suskiewicz, T. S., **J. E. K. Byrnes,** R. S. Steneck, R. Russell, C. J. Wilson, and D. B. Rasher. 2024. Ocean warming undermines the recovery resilience of New England kelp forests following a fishery-induced trophic cascade. Ecology:e4334. https://doi.org/10.1002/ecy.4334

59. Rosenthal, I. S.†, M. N. Simon, L. Trouille, and **J. E. K. Byrnes.** 2024. A citizen science approach to teaching climate change in introductory-level undergraduate general science courses. Journal of Geoscience Education 0:1–12. https://www.tandfonline.com/doi/full/10.1080/10899995.2023.2292945

58. **Byrnes, J. E. K**., A. Brown, K. Sheridan, T. Peller, J. Lawlor, J. Beaulieu, J. Muñoz, A. Hesketh, A. Pereira, N. S. Knight, L. Super, E. K. Bledsoe, J. B. Burant, J. A. Dijkstra, and K. Benes. 2024. Notes from the past show how local variability can stymie urchins and the rise of the reds in the Gulf of Maine. Ecosphere 15:e4800. https://doi.org/10.1002/ecs2.4800

57. Roy, M. S.†, **J. E. K. Byrnes,** and G. Mavrommati. 2023. Mitigation policies buffer multiple climate stressors in a socio-ecological salt marsh habitat. Sustainability Science. https://link.springer.com/article/10.1007/s11625-023-01414-0

56. Shaughnessy, B.K.†, Jackson, B., **Byrnes, J.E.K.** 2023. Evidence of elevated heavy metals concentrations in wild and farmed sugar kelp (*Saccharina latissima*) in New England. Scientific Reports. 17644. https://www.nature.com/articles/s41598-023-44685-4

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54. Eger, A. M., E. M. Marzinelli, R. Beas-Luna, C. O. Blain, L. K. Blamey, **J. E. K. Byrnes,** P. E. Carnell, C. G. Choi, M. Hessing-Lewis, K. Y. Kim, N. H. Kumagai, J. Lorda, P. Moore, Y. Nakamura, A. Pérez-Matus, O. Pontier, D. Smale, P. D. Steinberg, and A. Vergés. 2023. The value of ecosystem services in global marine kelp forests. *Nature Communications.* 14:1894. https://www.nature.com/articles/s41467-023-37385-0

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50. Ganley, L. C., **Byrnes, J.E.K.,** Pendleton, D. E., Mayo, C. A., Friedland, K. D., Redfern, J. V., Turner, J. T. and Brault, S. 2022. Effects of changing temperature phenology on the abundance of a critically endangered baleen whale. Global Ecology and Conservation 38: e02193. https://doi.org/10.1016/j.gecco.2022.e02193

49. Houskeeper H.F., Rosenthal I.S.†, Cavanaugh K.C., Pawlak C., Trouille L., **Byrnes J.E.K.,** Bell, T.W., and Cavanaugh, K.C.. 2022. Automated satellite remote sensing of giant kelp at the Falkland Islands (Islas Malvinas). PLoS ONE 17(1): e0257933. https://doi.org/10.1371/journal.pone.0257933

48. Hensel, M.J.†, Silliman, B.R., Hensel, E. and **Byrnes, J.E.K.** 2021. Feral hogs control brackish marsh plant communities over time. Ecology. Accepted Author Manuscript e03572. https://doi.org/10.1002/ecy.3572

47. Hensel, M. J. S.†, B. R. Silliman, J. van de Koppel, E. Hensel, S. J. Sharp, S. M. Crotty, and J. E. K. **Byrnes. 2021.** A large invasive consumer reduces coastal ecosystem resilience by disabling positive species interactions. Nature Communications 12:6290. https://doi.org/10.1038/s41467-021-26504-4

46. O'Connor, M. I., Mori, A S., Gonzalez, A., Dee, L. E., Loreau, M, Avolio, M., **Byrnes J. E. K.,** Cheung, W., Cowles, J., Clark, A. T., Hautier, Y., Hector, A., Komatsu, K., Newbold, T., Outhwaite, C. L., Reich, P. B., Seabloom, E., Williams, L., Wright, A. and Isbell, F. 2021. Grand challenges in biodiversity–ecosystem functioning research in the era of science–policy platforms require explicit consideration of feedbacks. *Proc. R. Soc. B.* 288202107832021078 3 http://doi.org/10.1098/rspb.2021.0783

45. Wong R.J.*, Roy M.S.†, **Byrnes J.E.K**. 2021 Sediment selection: range-expanding fiddler crabs are better burrowers than their historic-range counterparts. Marine Ecology Progress Series. 674:163-171. https://doi.org/10.3354/meps13811

44. Hautier, Y., Zhang, P., Loreau, M. WIlcox, K..R., Seabloom, E.W., Borer, E.T., **Byrnes, J.E.K**., *et al.* 2020. General destabilizing effects of eutrophication on grassland productivity at multiple spatial scales. *Nature Communications.* **11:** 5375 . https://doi.org/10.1038/s41467-020-19252-4

43. Lyman, T. P.†, Elsmore, K., Gaylord, B., **Byrnes, J. E. K**., & Miller, L. P. 2020. Open Wave Height Logger: An open source pressure sensor data logger for wave measurement. Limnology and Oceanography: Methods. https://doi.org/10.1002/lom3.10370 [website for logger]

42. Kirshen, P., Borrelli, M., **Byrnes, J.**, Chen, R., Lockwood, L., Watson, C., Starbuck, K., Wiggin, J., Novelly, A., Uiterwyk, K., Thurson, K., McMann, B., Foster, C., Sprague, H., Roberts, H. J., Bosma, K., Jin, D., & Herst, R. 2020. Integrated assessment of storm surge barrier systems under present and future climates and comparison to alternatives: A case study of Boston, USA. Climatic Change.https://doi.org/10.1007/s10584-020-02781-8

41. Chai, F., Xu, W., Musoke, T., Tarabelsi, G., Assaad, S., Freedman, J., Peterson, R., Piotrowska, K., **Byrnes, J.,** Rogers, S., Veraksa, A. 2019. Structure-function analysis of beta-arrestin Kurtz reveals a critical role of receptor interactions in downregulation of GPCR signaling in vivo. Developmental Biology. 455(2):209-419. Developmental Biology, 455(2), 409-419. http://dx.doi.org/10.1016/j.ydbio.2019.07.013

40. Blowes, S. A., S. R. Supp, L. H. Antão, A. Bates, H. Bruelheide, J. M. Chase, F. Moyes, A. Magurran, B. McGill, I. H. Myers-Smith, M. Winter, A. D. Bjorkman, D. E. Bowler, **J. E. K. Byrnes**, A. Gonzalez, J. Hines, F. Isbell, H. P. Jones, L. M. Navarro, P. L. Thompson, M. Vellend, C. Waldock, and M. Dornelas. 2019. The geography of biodiversity change in marine and terrestrial assemblages. Science 366:339.https://science.sciencemag.org/content/366/6463/339

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Steinberg, P.D., Stuart-Smith, R., Unsworth, R.K.F., van Keulen, M., Van Tussenbroek, B.I., Wang, M., Waycott, M., Weatherdon, L.V., Wernberg, T., Yaakub, S.M., 2019. Toward a coordinated global observing system for seagrasses and marine macroalgae. Front. Mar. Sci. 6. [link]

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34. O'Connor, M. I., Gonzalez, A., **Byrnes, J. E. K.**, Cardinale, B. J., Duffy, J. E., Gamfeldt, L., Griffin, J. N., Hooper, D., Hungate, B. A., Paquette, A., Thompson, P. L., Dee, L. E. and Dolan, K. L. 2017., A general biodiversity–function relationship is mediated by trophic level. Oikos, 126: 18–31. http://dx.doi.org/10.1111/oik.03652

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environment interactions over 500 million years. Global Change Biology. 21: 3595–3607. http://dx.doi.org/10.1111/gcb.12963

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14. Wheat, R. E., Wang, Y., **Byrnes, J. E.**, Ranganathan., J. 2013. Raising money for scientific research through crowdfunding. *Trends in Ecology & Evolution*. 28:71-72. http://dx.doi.org/10.1016/j.tree.2012.11.001

13. Hooper, D.U., Adair, E.C., Cardinale, B.J., **Byrnes, J.E.K.**, Hungate, B.A., Matulich, K.L., Gonzalez, A., Duffy, J.E., Gamfeldt, L., O'Connor, M.I. 2012. Biodiversity loss ranks as a major driver of ecosystem change. *Nature*. 286: 105-108. http://dx.doi.org/10.1038/nature11118

12. Ponette-González, A.G. and **Byrnes, J.E.** 2011. Sustainable science? Reducing the carbon impacts of mega-meetings. *Ethnobiology Letters*. 2:65-71. http://ethnobiology.org/publications/ebl/ethnobiology-letters-volume-2-2011/sustainable-science-reducing-carbon-impact-scien

11. **Byrnes, J.E.,** Reed, D.C., Cardinale, B.J., Cavanaugh, K.C., Holbrook, S.J., and Schmitt, R.J. 2011. Climate driven increases in storm frequency simplify kelp forest food webs. *Global Change Biology*. 17: 2513-2524. http://dx.doi.org/10.1111/j.1365-2486.2011.02409.x

10. Cardinale, B.J., Matulich, K., Hooper, D.U., **Byrnes, J.E.**, Duffy, E., Gamfeldt, L., Balvanera, P., O'Connor, M.I., Gonzalez, A. 2011. The functional role of producer diversity in ecosystems. *American Journal of Botany*. 98: 572-592. http://dx.doi.org/10.3732/ajb.1000364

9. Yang, L.J., Edwards, K.E., **Byrnes, J.E.,** Bastow, J.L., Wright, A.N., and Spence, K.O. 2010. A meta-analysis of resource pulse-consumer interactions. *Ecological Monographs*. 80: 125-151. http://dx.doi.org/10.1890/08-1996.1

8. **Byrnes, J.E.** and Stachowicz, J.J. 2009. The consequences of consumer diversity loss: different answers from different designs. *Ecology*. 90: 2879-2888. http://dx.doi.org/10.1890/08-1073.1

7. **Byrnes, J.E.** and Stachowicz, J.J. 2009. Short and Long Term consequences of increases in exotic species richness on water filtration by marine invertebrates. *Ecology Letters*. 8: 830-841. http://dx.doi.org/10.1111/j.1461-0248.2009.01339.x

6. Hughes A.R., **Byrnes J.E.**, Kimbro D.L. & Stachowicz J.J. 2007. Reciprocal relationships and potential feedbacks between biodiversity and disturbance. *Ecology Letters*. 10: 849-864. http://dx.doi.org/10.1111/j.1461-0248.2007.01075.x

5. **Byrnes, J.E.**, Reynolds, P.L., Stachowicz, J.J. 2007. Invasions and extinctions reshape coastal marine food webs. *PLoS One*. 2: e295. http://dx.doi.org/10.1371/journal.pone.0000295

4. Bullard S. G., Lambert G, Carman M.R., **Byrnes J.,** Whitlatch R.B., Ruiz G., Miller R.J., Harris L., Valentine P.C., Collie J.S., Pederson J., McNaught D.C., Cohen A.N., Asch R.G., Dijkstra J., Heinonen K. 2007. The invasive colonial ascidian *Didemnum* sp.: current distribution, basic biology, and potential threat to marine communities of the northeast and west coasts of North America. *Journal of Experimental Marine Biology and Ecology*. 342: 99-108. http://dx.doi.org/10.1016/j.jembe.2006.10.020

3. Stachowicz, J.J. and **Byrnes, J.E.** 2006. Species diversity, invasion success and ecosystem

functioning: combining experimental and observational approaches to assess the roles of resource competition, facilitation and extrinsic factors. *Marine Ecology Progress Series*. 311: 251-262. http://dx.doi.org/10.3354/meps311251

2. **Byrnes, J.E**., Stachowicz, J.J., Hultgren, K.M., Hughes, A.R., Olyarnik, S.V., Thornber, C. 2006. Predator Diversity Enhances Trophic Cascades in Kelp Forests by Modifying Herbivore Behavior. *Ecology Letters*. 9: 61-71. http://dx.doi.org/10.1111/j.1461-0248.2005.00842.x

1. **Byrnes, J.E.** and Witman, J.D. 2003. Impact assessment of an invasive flatworm, *Convoluta convoluta*, in the southern Gulf of Maine. *Journal of Experimental Marine Biology and Ecology*. 293: 173-191. http://dx.doi.org/10.1016/S0022-0981(03)00166-7

Additional Preprints

1. **Byrnes J.E.K**, Johnson L.E., Connell S.D. et al. 2013. The sea urchin – the ultimate herbivore and biogeographic variability in its ability to deforest kelp ecosystems. *PeerJ PrePrints*, 1, e174v1. http://dx.doi.org/10.7287/peerj.preprints.174v1

2. Dunic, J.C.†, Elahi, R., Hensel, M.J.S.†, Kearns, P.J., O'Connor, M.I., Acuña-Hurtado, D., Honig, A., Wilson, A.R., **Byrnes, J.E.K.**, 2020. Human activities influence the direction and magnitude of local biodiversity change over time. *bioRxiv* 162362. http://dx.doi.org/10.1101/162362

3. **Byrnes, J.,E.K.** Dissly, L., Bosma, K. 2022. Surprising Abundant Mussel Beds in the Center of Boston Harbor in the Midst of a Regional Die-Back. *EcoEvoRxiv*. https://ecoevorxiv.org/repository/view/3784/

4. Mantha, K. B., R. Sankar, Y. Zheng, L. Fortson, T. Pengo, D. Mashek, M. Sanders, T. Christensen, J. Salisbury, L. Trouille, **J. E. K. Byrnes**, I. Rosenthal, H. Houskeeper, and K. Cavanaugh. 2022, November 7. From fat droplets to floating forests: cross-domain transfer learning using a PatchGAN-based segmentation model. arXiv. https://arxiv.org/abs/2211.03937

Book Chapters

3. O'Connor, M.I. and **Byrnes, J. E.K.** Biodiversity and Ecosystem Function in Marine Ecosystems. 2014. In Marine Community Ecology and Conservation, M. Bertness, J. Bruno, B. Silliman, and J. Stachowicz, eds.

2. Fox, J., **Byrnes, J.**, Boker, S., and Neale, M. 2011. Structural equation modeling in R with the *sem* and *OpenMX* packages. In *Handbook of Structural Equation Modeling*. Rick H. Hoyle, David Kaplan, George Marcoulides, and Steve West, eds.

1. Olyarnik, S.V., Bracken, M.E., **Byrnes, J.E.**, Hughes, A.R., Hultgren, K.M., Stachowicz, J.J. 2009. Ecological factors affecting community invasibility. In *Biological Invasions in Marine Ecosystems: Ecological, Management, and Geographic Perspectives*. Gil Rilov and Jeff Crooks, eds. Springer-Verlag. pp. 215-138.

Reports

4. Global Ocean Observing System. 2024. Essential Ocean Variable Specification Sheet: Macroalgal canopy cover and composition. https://goosocean.org/document/17515
*- Co-Lead of panel developing specification sheet.

3. Intergovernmental Panel on Biodiversity and Ecosystem Services. 2019. Chapter 4: Plausible futures of nature, its contributions to people and their good quality of life. In IPBES Global Assessment on Biodiversity and Ecosystem Services (Deliverable 2c). Government draft to be submitted for approval at the IPBES 7th Plenary, May 2019, Paris. https://www.ipbes.net/document-library-catalogue/chapter-4-plausible-futures-nature-its-contributions-people-and-their
*-Contributing authors are not listed in main citation as per IPBES policy

2. Cavender-Bares, J., Arroyo, M.T.K., Abell, R., Ackerly, D., Ackerman, D.,Arim, M., Belnap, J., Castañeda Moya, F., Dee, L., Estrada-Carmona, N., Gobin, J., Isbell, F., Köhler, G., Koops, M., Kraft, N., Mcfarlane, N., Martínez-Garza, C., Metzger, J. P., Mora, A., Oatham, M., Paglia, A., Pedrana, J., Peri, P. L., Piñeiro, G., Randall, R., Robbins, W. W., Weis, J., and Ziller, S. R. 2018. Chapter 3 Status and trends of biodiversity and ecosystem functions underpinning nature's benefit to people. In: The IPBES regional assessment report on biodiversity and ecosystem services for the Americas. Rice, J., Seixas, C. S., Zaccagnini, M. E., Bedoya-Gaitán, M., and Valderrama, N. (eds.). Secretariat of the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services, Bonn, Germany. https://www.ipbes.net/assessment-reports/americas
*-Contributing authors are not listed in main citation as per IPBES policy

1. Kirshen, P., K. Thurson, B. McMann, C. Foster, H. Sprague, H. Roberts, M. Borelli, **J. E. K. Byrnes**, R. Chen, L. Lockwood, C. Watson, K. Starbuck, J. Wiggin, A. Novelly, K. Ulterwyk, K. Bostoma, E. Holmes, Z. Stromer, J. Famely, A. Shaw, B. Hoffnagle, Di Jin, R. Herst, E. Moothart, and C. Humphries. 2018. Feasibility of Harbor-wide Barrier Systems. Sustainable Solutions Lab, University of Massachusetts Boston. https://scholarworks.umb.edu/environment_pubs/4/

Software:
piecewiseSEM - Analysis of structural equation models using local area estimation techniques in R. Co-author/contributor as of version 2.0. http://jslefche.github.io/piecewiseSEM/

multifunc - Analysis of simultaneous change in multiple ecosystem functions in R. http://github.com/jebyrnes/multifunc

lavaan - Analysis of latent variable structural equation models in R. Contributing Developer. http://lavaan.org

Films

Fear and Fishing in Lake Davis. Documentary film produced in collaboration with 5 other UC Davis students over two years detailing the invasion of Northern Pike in California and the politics of invasive species eradication efforts. Lead Editor. Flag in the Ground Productions. http://cpb.ucdavis.edu/bioinv/projects/pike/. 2005.

Online

2016 - 2020. Deep Sea News. http://deepseanews.com Ocean science outreach blog. Contributor.

2014 - 2023. Floating Forests Blog. http://blog.floatingforests.org/ Outreach and communications portion of online citizen science project to discuss kelp forest biology and ecology.

2002-Present. I'm a chordata! Urochordata! http://www.imachordata.com/. A science blog discussing ecology, marine biology, and the culture of science in the modern age.

2007. Embracing Blogs and Other Tools of the Information Age. http://www.esa.org/esablog/?p=21. An article for Ecotone, the Ecological Society of America blog.

Online Projects:

2014 - 2023. Floating Forests. http://www.floatingforests.org/ Online citizen science project to identify trends in global kelp abundance using satellite imagery.

2011 - 2023. The #SciFund Challenge. A large-scale effort to attempt to get scientists to crowdfund their research. https://experiment.com/institutions/scifund and http://scifundchallenge.org/ for projects and for the project blog. Project raised $307,825 in 4 rounds. 182 scientists participating. Six publications from projects as of 8/1/2014.

Personnel Mentored & Current Positions:
For most up to date list and where they are, see https://celp-lab.org/lab_alums.html

UMass Boston Postdocs: Alison Haupt (Assistant Professor, California State University Monterey Bay), Christian Alsterberg (current Environmental Coordinator på Catchment Partnership of Kävlingeån), Meredith McPherson (USGS), Sundarabalan. V. Balasubramanian.

UMass Boston PhD Students: Marc Hensel (Research Assistant Professor at the University of Florida), Breckie McCollum (Assistant Professor at Berklee College of Music), Michael Roy (Data Science Lead for the New York State Department of Public Health), Brianna Shaughnessy (NOAA Communications Specialist with the Fisheries Office of Communications and Seafood Literacy Coordinator), Issac Rosenthal (Postdoc at Amherst College), Alexis Wilkes, Daniel Lopez, Curtis Morris, Melba Torres Sosa

UMass Boston Masters Students: Jillian Dunic (Staff at the Department of Fisheries and Oceans, Canada), Ted Lyman (completed degree), Lucy Lockwood (Current PhD Student at UMass Boston with Paul Kirshen), Wendi White (Lead technician in Hughes Lab at Northeastern University)

UMass Boston Undergraduate: Amy Jones, Melody Rose, Michael Moscote, Richard Wong (PhD Student at Duke University), Linnea Sturdy (Masters Student at University of New England), Farah Ahmad (Community Impact Programs and Operations Coordinator at United Way of Western CT), Nour Tabidi, Chena Farhat (Law student at Suffolk University), Sarah Strohacker, Nelson Nease, Felicia Horton, Melody Rose, Taylor Kielczewski (PhD Student University of Auckland)

REU and Research Interns. Madison Lin, Amy Smith, Jillian Hubbard, Alannah Dodd, Laura Dissly, Lianna Greenberg-Nielsen, Sarah McCormak, Katie Bland (Masters student, School of Marine and Environmental Affairs, University of Washington), Conner Donovan (Massachusetts Water Resources Authority), Jeremiah Ets-Hokin (Masters student at San Francisco State University)

UCSB: Jocelyn Enevoldsen (Researcher, Cal Poly Humboldt), Eric Dilley (Rapid Response Coordinator @ Research Corp. of UH | Marine Biology), Matt Foster (Finance and Operations at Black River Caviar)

UC Davis: Jonathan Schram (Essential Fish Habitat Consultation Coordinator at Ocean Associates, Inc.), Cameron Coates (Scientist at Zymergen, Inc.), Tanya Rogers (Research Fish Biologist at NOAA)

Teaching:

2024. COBALT Casco 2024 GIS in R Workshop. COBALT. Portland, ME.

2023, 2024. Structural Equation Modeling for Ecology and Evolutionary Biology. P.R. Statistics.

2019 - Present. Underwater Research. Shoals Marine Laboratory.

2017. Disentangling Complex Causal Relationships in Spatial and Temporal data with SEM. University of Tromsø.

2017. Disentangling Complex Causal Relationships in Spatial and Temporal data with SEM. University of Tromsø.

2017-Present. BIOL609: Advanced Data Analysis for Biology. UMass Boston. http://biol609.github.io

2016-Present. Marine Conservation Biology. Shoals Marine Laboratory.

2016-Present. BIOL355: Introduction to Data Science for Biology. UMass Boston. http://biol355.github.io

2014-Present. BIOL310: Marine Biology. UMass Boston.

2014. BIOL697: Introduction to Meta-analysis for Biology. UMass Boston

2012-Present. BIOL607: Biostatistics and Experimental Design. UMass Boston. http://biol607.github.io

2012. An Introduction to Structural Equation Modeling - a Tool for Studying Complex Systems. Co-Taught with James Grace. Stockholm University, Sweden.

2011-Present. Using tools from Theater for Science Presentations. Co-taught with Lila Rose Kaplan, International Science Playwriting award winner. NCEAS. Kavli Institute for Theoretical Physics. UMass Boston.

2010-Present. Structural Equation Modeling for Ecologists and Evolutionary Biologists. 3-5 day Workshop. Lovén Center for Marine Sciences. University of Zurich. National Center for Ecological Analysis and Synthesis. Universidad Nacional Autónoma de México, Morelia, University of British Columbia. Virginia Institute of Marine Science, University of Coimbra. UMass Boston, UNC Chapel Hill. University of Swansea. PR Statistics. Bodega Marine Lab, University of Lund http://byrneslab.net/teaching/sem/

Grants and Awards:

2026 – 2027. Democratizing Seaweed Knowledge: Creating an Open-Access Digital Repository to Facilitate Knowledge Transfer and Stimulate Innovation in RI. Rhode Island STAC. Subrecipient. $21,100

2026.Building a Demonstration Kelp Hatchery at Cat Cove Marine Lab. Massachusetts Department of Marine Fisheries. $49,200

2026. Building Tools to Map Kelp Beds in Massachusetts Coastal Waters. Massachusetts Department of Marine Fisheries. $57,543

2025. Stone Living Lab. Co-PI. James M. and Cathleen D. Stone Foundation.

2024. Stone Living Lab: Living Seawalls. Massachusetts Bays National Estuarine Partnership. $16,089.

2024. Stone Living Lab: ACS. James M. and Cathleen D. Stone Foundation. $124,330

2024. Stone Living Lab: Living Seawalls. James M. and Cathleen D. Stone Foundation. $116,889

2023. Stone Living Lab: Living Seawalls. James M. and Cathleen D. Stone Foundation. $102,014

2023. Stone Living Lab: Data Management. James M. and Cathleen D. Stone Foundation. $127,346.

2021-2024. Vulnerability of giant kelp populations to climate change. NASA. $169,222 as Co-I.

2021 - 2024: Using Hyperspectral Imagery to Assess the Effects of Warming on New England Kelp Forests. NASA MUREP Program. $742,227.

2021 - 2022. Gathering crucial baseline data for multiple sectors to address knowledge gaps in *Saccharina latissima* (sugar kelp) distribution and environmental services along Massachusetts coasts. UMass Boston Proposal Development Grant. $19,350.

2021-2022 Detecting and Attributing Synchrony in Cross-Ecosystem Community Dynamics Using Long-Term Observational Data in Coastal Systems. Canadian Institute for Ecology and Evolution. Funding for two virtual workshops.

2020 - 2023. The Stone Living Laboratory. Stone Foundation. $383,422 as Co-I

2018 - 2022. Renewal: Using Citizen Science to Understand Thirty Years of Change in Global Kelp Cover by Expanding the Zooniverse to NASA Satellite Imagery. NASA. $804,557 as lead PI.

2017 - 2018. Using Citizen Science to Understand Thirty Years of Change in Global Kelp Cover by Expanding the Zooniverse to NASA Satellite Imagery. NASA. $163,972 as lead PI.

2016 - 2018. Improving the environment while protecting our coasts: A holistic accounting of ecosystem services of "Green Infrastructure and Natural and Nature-Based Features (NNBF)" in an urbanized coastal environment. NOAA. $10,707 as a Co-PI.

2016-2018. Support for student marine research efforts. MPH Foundation. $36,000 as PI over multiple years.

2015 - 2022. LTER-Plum Island Ecosystems: Dynamics of Coastal ecosystems in a region of rapid climate change, sea-level rise, and human impacts. National Science Foundation. $360,000 as a contractor.

2016-2018. Evaluating the relationship between kelp forests ecosystems and water temperature in the southern Gulf of Maine. Woods Hole Sea Grant. $100,119

2015-2017. Feedbacks Between Coastal New England Kelp Beds and Wave Disturbance. MIT SeaGrant. $150,000

2015-2016. Changes in the Importance of Bird Predation in the Face of Climate Driven Range Shifts in New England. UMB Healey Grant. $7,500

2014-2016. Food Web Structure as a Driver of Multiple Ecosystem Functions in new England Salt Marsh Ecosystems. MIT SeaGrant. $165,536

2014-2015. The Role of Large-Scale Kelp Die-Offs in Facilitation of Invasive Species. UMB Healey Grant. $7,500

2011. Hey! Where's that fish?! A #SciFund Crowdfunded project. $4,600.

2010-2012. Nation Center for Ecological Analysis and Synthesis, Postdoctoral Fellowship. Full Stipend.

2007. Western Society of Naturalists Student Travel Award. $500.

2006. Achievement Rewards for College Scientists (ARCS) Fellow. Full Stipend.

2002-2004, 2007. NSF Integrative Graduate Education and Research Traineeship Long Term Fellowship in Biological Invasions. Full Stipend.

2003-2004. UC Davis Center for Population Biology Research Fellowship. $2500.

2001. Susan Colver Rosenberger Prize in Biology.

2000. NSF Research Experience for Undergraduates. $2000.

2000. Brown Undergraduate Teaching and Research Assistantship. $2000.

Synergistic Activities:

Working Groups & Organizations

2026. Improving Predictions in Ecology. Andina. Nerín, Spain.

2026. Drivers and trends in NW Atlantic kelp forests. Canadian Institute for Ecology and Evolution. Co-Lead. Quebec City.

2024. Kelp species distributions in the Northwest Atlantic. DFO. Halifax, Nova Scotia.

2023. Causality in Ecology. Johns Hopkins University working group.

2021 - 2022. Detecting and Attributing Synchrony in Cross-Ecosystem Community Dynamics Using Long-Term Observational Data in Coastal Systems. Canadian Institute for Ecology and Evolution. Virtual meetings.

2019. Macroalgal Essential Ocean Variable (EOV) data processing and workflow. Co-Hosted by GOOS and the ARDC. IMAS. Hobart, Australia.

2018. Planning the implementation of a global long-term observing and data sharing strategy for macroalgal communities. Co-Hosted by the Global Ocean Observing System and the Partnership for Observation of the Global Ocean. IMAS. Hobart, Australia.

2017-2019. Scaling Up Productivity Responses to Changes in Biodiversity. NCEAS LTER working group.

2016. Quantifying Biodiversity Change through Time. sDiv working group.

2015. Detection and Attribution of Biodiversity Change. CIEE working group.

2014-Present. Floating Forests Science Team. http://floatingforests.org.

2014-Present. Kelp Ecosystem Ecology Network. Network co-ordinator.

2014-Present. Director of the Board, The #SciFund Challenge. A non-profit dedicated to training scientists to use crowdfunding as a tool for outreach.

2013. Global Impacts of Climate Change on Kelp Forests. Leader, National Center for Ecological Analysis and Synthesis working group.

2012-2013. The future of publishing in ecology, evolution, and environmental sciences. Leader, National Center for Ecological Analysis and Synthesis working group.

2011-2012. Determinants of extinction in ancient and modern seas. National Evolutionary Synthesis Center working group.

2010-2011. Biodiversity and the functioning of ecosystems: translating results from model experiments to functional reality. National Center for Ecological Analysis and Synthesis working group.

2010. Dissertation Initiative for the advancement of Climate Change ReSearch (DISCCRS) participant. Interdisciplinary workshop in climate change communication.

Outreach

2017. The Kelp Ecosystem Ecology Network of New England. Boston Sea Rovers.

2014. The Biology of our Oceans in a Changing World. Cohasset Student Center for Coastal Research.

2011. Statistics and the Natural World. Alameda High School.

2011. Kelp Forest Research. Palo Alto Jewish Community Center.

2011. Kids do Ecology. Helping 4th graders conduct their own ecological experiments and present results to their peers. Topic: Does light reduce kelp consumption by beach hoppers?

2010-2011. Speaker, Santa Barbara Coastal LTER High-School Teacher Ocean Science Workshop.

2009-2011. Editor at Researchblogging.org.

2004-2005. Intern, Gulf of the Farallones National Marine Sanctuary and Cordell Bank National Marine Sanctuary. Created species list of all non-native species either present in sanctuary waters, or likely to invade in the near future.

Professional Service

2026-2027. President. Benthic Ecology Meeting Society.

2025-2026. President-Elect. Benthic Ecology Meeting Society.

2025-Present. Member, International Union for Conservation of Nature Seaweed Specialist Group.

2024-Present. Co-Chair, Macroalgal Community Cover and Composition EOV group as part of the Global Ocean Observing System Bio-Eco Panel.

2024. Panelist, NASA Biodiversity and Ecological Conservation Researching Scale Subelement grants review panel.

2022. Panelist, NSF Biological Oceanography grants panel.

2022 - 2024. Committee Member. Benthic Ecology Meeting Society.

2021 - 2025.. Handling Editor. *Journal of Experimental Marine Biology and Ecology.*

2016. Panelist, NCEAS LTER synthesis grant reviews.

2013 - Present. Ad hoc Grant reviewer for NSF, NSERC

2013 - 2025. Handling Editor. *Oikos*

2013 - 2019. Handling Editor: *Evolution & Ecology,*

2013 - 2016. Handling Editor. *Axios Reviews*

2011. Contributor, Faculty of 1000. http://goo.gl/ekVCz

2010 - 2014. Editor, Ecology on the Web. *Bulletin of the Ecological Society of America.*

Reviewer. *Nature Communications*, *Ecology*, *Ecological Monographs*, *Ecology Letters*, *Journal of Animal Ecology*, *Oecologia*, *Oikos, Evolution & Ecology, PLoS One, Marine Ecology Progress Series, Botanica Marina*, *Hydrobiologia.*

Western Society of Naturalists Student Committee. 2005-2007. Organized 2005 symposium on Cross Ecosystem interactions. Organized symposium "The Natural History of the Channel Islands" in 2007. Created and maintained online meeting registration and submission system.

Invited Presentations:

2026. Living Seawalls in Massachusetts: 8 Month Biodiversity Report. Stone Living Lab Seminar Series.

2026. Living Seawalls in Massachusetts: If You Build It, Who Will Come? NorthEast Algal Society (NEAS).

2024-2025. Living Seawalls in Massachusetts. Salem Sound Coastwatch, New England Aquarium Dive Club, Sasaki.

2023. Understanding Drivers of Ecological Change by Applying Causal Analysis to Observational Data. Boston University

2022. Kelp Forests of New England in an Era of Global Change, Roger Williams University

2021. Understanding Change in Kelp Forests by Applying Causal Analysis to Observational Data, UMass Amherst.

2021. OOPS! I forgot to measure that! Coping with omitted variable bias for the causal analysis of observational data. Ecological Society of America.

2020. Macroalgal (and community) Benthic Monitoring: Can't We All Just Get Along?, Hakai Institute.

2020. Understanding Changes in Kelp Forests Around the World and Around Appledore, Rock Talk Seminar Series, Shoals Marine Lab.

2020. From kelp foRests to coRonavirus an inteRactive jouRney in doing science with R, University of Rhode Island R Studygroup.

2019. Global Data for Global Kelp Forest Monitoring, Parternship for Global Oceans Macroalgal EOV Meeting, POGO, Hobart, Tasmania.

2018-2019. Connecting Global Change to Shifts in Biodiversity and Ecosystem Function in Coastal Ecosystems. Global change in marine ecosystems: from foundation species to biodiversity. University of Maine. University of Massachusetts Boston. Texas A&M Corpus Christi

2018. Challenges in understanding kelp forest dynamics at the scale of the planet. Gordon Conference on Ocean Global Change Biology.

2016-2018. Global change in marine ecosystems: from foundation species to biodiversity. University of Maine. University of Connecticut. Bigelow Labs. Bodega Marine Lab

2015. Tools from theater for scientists. UMass Boston Coasts and Communities IGERT seminar with Lila Rose Kaplan.

2015. The future of floating forests in a changing world. Southern Connecticut State College.

2015. Connecting your trainees and visitors to ocean science onshore! National Tall Ships Association Conference.

2014-2016. Of Kelp and Climate Change: Reef Life in your Back Yard. New England Aquarium. Monadnock Summer Lyceum. Salem Sound Coastwatch.

2014. Salt marsh surveys. Nantucket Biodiversity Initiative.

2014. Diversity is the spice of (ocean) life. Cohasset Center for Student Coastal Research.

2014. Of indirect effects, climate change, and crowdfunding science. Woods Hole Oceanographic Institute.

2014. Global change, species diversity, and the future of marine ecosystems. Marine Biological Laboratory.

2013. Crowdfunding Science: Appealing to the online community for research money. American Academy for the Advancement of Science webinar.

2013. Measuring the effects of change in biodiversity on multiple ecosystem functions. Multifunctional landscapes symposium. International Association for Ecology (INTECOL) Congress.

2013. Causes and consequences of ecological complexity. Brown University, Dalhousie University.

2012. Taking the Ecological Conversation Online. Ecological Society of America.

2012. Engagement leads to science funding: the #SciFund Challenge. NCEAS, Coral Reef Alliance.

2011. Consequences of climate driven shifts in kelp forest food web complexity. Romberg Tiburon Center for Environmental Studies. Hopkins Marine Lab

2011. Beyond species richness: food web structure and ecosystem function. World Conference on Marine Biodiversity.

2011. Causes and consequences of ecological complexity. Stony Brook University. Northeastern University. Universidad Nacional Autónoma de México, Morelia.

2010. Giving a great talk: How to use your body and your voice. With Lila Rose Kaplan and Anne Torsiglieri. Kavli Institute for Theoretical Physics.

2009, 2010. Causes and consequences of biodiversity in southern Californian kelp forests. National Center for Ecological Analysis and Synthesis. San Diego State University Ecology and Evolutionary Biology Seminar Series. University of California Santa Barbara EEMB Seminar Series.

2008. Consequences of declining predator diversity in the ocean: a tale of two systems. Brown University Conservation Biology Seminar Series.

2008. Using structural equation modeling for marine ecological data. Subtidal Ecology Class. Dr. Michael Graham. Moss Landing Marine Laboratory.

2007. The consequences of changing biodiversity in marine ecosystems. California State University, Stanislaus.

2007. Combining long-term surveys with structural equation modeling to examine kelp forest food webs. North American Benthological Society Meetings. Invited for the symposium Cascades, Chains, and Webs: Alternative Approaches to Understanding Complex Systems"

2007. Identifying the ascidians of San Diego. Scripps Institute of Oceanography.

2005. Predator diversity enhances trophic cascade strength through changes in herbivore behavior. Ecological Society of America Annual Meeting. Invited for the symposium *Incorporating trophic diversity into the biodiversity-ecosystem functioning debate: Lessons from aquatic ecosystems*.

Contributed Presentations:

2025. Environmental Change Drives Declines in Kelp Abundance Across the Northwest Atlantic, International Temperate Reefs Symposium.

2024. Mapping sugar kelp (*Saccharina latissima*) on a subsurface temperate reef system in the optically complex water column of Salem Sound, Massachusetts. Benthic Ecology Meeting.

2023. Oh S**t! I forgot to measure that! Coping with omitted variable bias for the causal analysis of observational data. Benthic Ecology Meeting. Western Society of Naturalists.

2023. Global Change with Forty Years of Data from Intertidal and Subtidal Reefs in Maine, International Temperate Reefs Symposium.

2023. Oh S**T! I forgot to measure that! Coping with omitted variable bias for the causal analysis of observational data, Benthic Ecology Meeting.

2022. Natural History Observations Show Wave Exposure Slowing Urchins and the Rise of the Reds in the Gulf of Maine, Benthic Ecology Meeting, Portsmouth, NH

2019. **Jarrett E.K. Byrnes** , Haupt A.J., Dijkstra J.A., Grabowski J.H., McMahan M., Rasher D., Humphries A.T., Thornber C. Temperature drives declines in kelp forests and biodiversity. Benthic Ecology Meeting. St. John's, Newfoundland, Canada.

2017. **Byrnes, J.E.K.,** Cavanaugh K.C., Haupt, A.J., Trouille, L., Rosenthal, I., Bell, T.W., Rassweiler, A., Pérez-Matus, A., and J. Assis. Using Online Citizen Science to Assess Giant Kelp Abundances Across the Globe with Satellite Imagery. American Geophysical Union.

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Professional Associations:

Benthic Ecology Meeting Society, Ecological Society of America, Western Society of Naturalists, British Ecological Society, American Association for the Advancement of Science, American Academy of Underwater Scientists

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